The Accountability Gap in Tesla's Autopilot and Full Self-Driving Systems: Marketing, Complacency, and the Ethical Obligations of Computing Professionals

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1. Introduction

In late 2015, Tesla introduced Autopilot, a partially automated system that is promising to reduce the workload and stress maintained by the drivers. Five years later, the company launched an extended version of autopilot, the Full-Self Driving System (FSD), a paid software extending the features first introduced in Autopilot and marketed as the pathway to fully autonomous, driverless operation.

By 2024, the United States National Highway Traffic Safety Administration (NHTSA) had recorded 956 crashes involving these systems, resulting in 29 fatalities (Gigazine, 2024). These figures do not merely represent technical failures; they are the measurable consequences of a widening gap between the capability Tesla marketed and the capability its systems actually possessed.

This report investigates that accountability gap: how Tesla's naming conventions, marketing language, and system design choices collectively induced driver complacency, enabled foreseeable misuse, and externalised fatal risk onto third parties who had no say in whether the technology was ready for public deployment.

Now we ask: ***How has Tesla's Autopilot and FSD system created an accountability gap between marketed capability and real-world safety performance, and what does this mean for the computing professionals who build and deploy safety-critical Artificial Intelligence (AI)?*** To answer it, this report examines the technical architecture of both systems, the documented pattern of incidents and recalls, the structural incentives that shaped product decisions, and the professional ethical frameworks that should have, but did not, govern those decisions.

This topic matters for computing professionals beyond legal liability. The engineers who removed radar from Tesla's sensing stack, the product team who named an unfinished system "Full Self-Driving," and the software team who deployed over-the-air updates without formal regulatory recall processes, each made professional decisions within institutional incentive structures that prioritised commercial ambition over public safety.

The Institute of Electrical and Electronics Engineers (IEEE) Code of Ethics (2020) and the Australian Computing Society (ACS) Code of Professional Conduct both require members to

place the safety of the public above other considerations. Tesla's case is, in this sense, a study in what happens when those obligations are systematically violated. The report proceeds through a literature review of the technology and its documented failures and a structured problem analysis using a HOW–WHAT–WHY framework.

2. Literature review

2.1 The Technology: How Autopilot and FSD Work

Tesla Autopilot, in its original form, combined camera, radar, and ultrasonic sensor data to support highway driving functions: Traffic-Aware Cruise Control, Autosteer, and automated lane changes. The system is classified as SAE Level 2 partial automation, which by definition means the human driver remains the operator at all times and must be ready to intervene immediately (SAE International, 2021). Autopilot's driver monitoring relies on a torque-based steering wheel sensor, a design choice that proved exploitable, as a documented user mentioned that "*Honestly, this is what I do most of the time. I'll actually have a little weight that you can clip on to the steering wheel that negates all the nagging that it does*." (R068, as cited in De Winter, J.C.F., et al., 2023).

FSD, available as a paid add-on priced between with a USD 99/month subscription, extends the Operational Design Domain (ODD) beyond highways to urban and non-highway environments (Korucuoğlu, 2025). It relies on an end-to-end deep learning architecture: eight cameras feed into a transformer-based neural network that produces real-time scene understanding, followed by path-planning networks that select vehicle actions using reinforcement learning principles (AutoRaiders, 2025). Tesla describes this training pipeline as processing millions of miles of real-world fleet data to enable the system to identify objects, infer velocities and trajectories, and navigate complex intersections (Tesla, n.d.).

A critical architectural decision occurred in 2021 when Tesla removed radar from its vehicles, transitioning to a camera-only system it branded "Tesla Vision." Internal accounts, later reported by The Washington Post (2023), indicate that Tesla engineers warned Elon Musk that removing radar would cause crashes in scenarios where cameras became obscured, by sun glare, rain, or

sensor contamination. Musk overruled those objections. The Washington Post investigation documented an immediate uptick in near-misses and crashes following the removal, a finding that transforms what might otherwise appear to be a technical limitation into an identifiable professional and ethical failure: a deliberate management decision made against the advice of the engineers responsible for public safety.

Neither Autopilot nor FSD, despite their names, constitutes a fully autonomous system. Tesla's own documentation acknowledges that a human must supervise and be prepared to take control at all times (Siberoloji, 2025). FSD was renamed from "FSD Beta" to "FSD (Supervised)" in April 2024, amid regulatory scrutiny and marketing backlash, a renaming that functions, in retrospect, as a corporate admission that the original branding was misleading (Teslarati, 2024).

2.2 Documented Incidents and the Pattern of Failure

The academic and journalistic literature identifies not isolated incidents but a systemic pattern of failure. The 2018 death of Walter Huang, a 38-year-old Apple engineer whose Tesla Model X veered into a concrete highway barrier near Mountain View, California, while Autopilot was engaged, established early that the system could not reliably detect stationary objects even in well-mapped highway corridors. Huang's hands were not on the wheel; NHTSA's investigation found Autopilot had been engaged for approximately 19 minutes at 71 mph before the crash (CBS News, 2022). The case raised an immediate question: to what degree was driver inattention a foreseeable consequence of the system's design and naming?

That question acquired greater urgency with the 2023 recall of 362,000 FSD Beta vehicles, after NHTSA determined that FSD Beta posed unreasonable safety risks by exceeding speed limits, driving through intersections unsafely, and misreading road conditions (CNET, 2023). The recall was addressed through an over-the-air software update, a process that, while efficient, bypassed the formal safety validation processes associated with traditional automotive recalls. This mechanism, deploying updates without regulatory approval, was itself a professional ethics issue: engineers and product teams were effectively conducting public beta tests on live roads.

The death of Jeffrey Nissen in 2024 illustrates the third-party dimension of these failures. A Tesla operating with FSD activated struck and killed Nissen, a 28-year-old motorcyclist, on State Route 522 in Snohomish County, Washington. The driver was looking at his phone at the time of

impact, a behaviour made possible by the trust the system's name and marketing had cultivated (AIAAIC, 2024). Nissen had never purchased a Tesla, never agreed to participate in an autonomous vehicle deployment, and had no mechanism to opt out of the risk. His death is emblematic of what Bainbridge (1983) identified decades ago as the "irony of automation": systems designed to reduce human error can, when poorly designed or misnamed, introduce new and more dangerous forms of human failure by undermining the vigilance the system still requires.

2.3 The Research Evidence on Complacency and Over-Trust

De Winter, J.C.F., et al. (2023) conducted the most comprehensive empirical investigation of Autopilot and FSD user behaviour to date, analysing 103 real Tesla drivers through semi-structured interviews. Their findings are direct and damning: drivers reported becoming complacent over time with Autopilot engaged, taking their eyes off the road for extended stretches, in one case, a driver reported not looking up for 44 miles, and engaging in secondary activities including laptop use. Rather than actively steering, drivers placed their hands on the wheel primarily to satisfy the torque sensor. The study also found that while Autopilot induced complacency, FSD induced stress, with users reporting that the system's unpredictability in urban environments required what one participant called "literally full attention all the time, I'm driving at 200% attention with FSD Beta" (R023, as cited in De Winter, J.C.F., et al., 2023).

Crucially, De Winter, J.C.F., et al. (2023) do not frame this complacency as a driver failure. They attribute it to the system's design and naming: the name "Autopilot" activated aviation-derived mental models of full automation; and the interface provided no meaningful cognitive scaffolding to remind drivers of the system's actual limitations. This analysis is substantiated by Chu and Liu (2023), who found that in crash investigations involving ADAS systems, driver inattention was a recurring causal factor, but that the vehicle's *operational design* permitted the inappropriate driver behaviour in the first place. The distinction is clinically important: it shifts accountability from individual users to the professional decisions embedded in the system's architecture and communication.

A further psychological dimension emerges from De Winter et al.'s data: prolonged use of Autopilot appears to degrade drivers' manual driving confidence. One participant reported, "As

soon as I have to take over, I feel very uncomfortable for some reason. I used to be very comfortable driving on my own" (R074, as cited in De Winter, J.C.F., et al., 2023). This finding points to a cascading harm — not only does the system induce complacency, but extended reliance on it may erode the very skill set that the SAE Level 2 classification assumes the driver still possesses.

2.4 Data Privacy: A Secondary Ethical Dimension

Tesla's camera-first architecture raises a parallel ethical concern: the privacy of non-consenting parties captured by the vehicle's cameras. Tesla's privacy policy states the company does not sell user data to third parties (Tesla, n.d.). However, a Reuters investigation (2023) found that employees had shared sensitive footage, including recordings of customers in private situations and footage captured inside people's garages, through internal messaging systems and private group chats, in some cases creating memes from the images. Former employees confirmed that while recordings were nominally anonymised, they often included location data sufficient to identify individual Tesla owners (Reuters, 2023).

This data was collected from people who had never consented to being recorded — pedestrians, neighbours, and bystanders captured incidentally by vehicles parked on public or private streets. The German Data Protection Authority, in a formal statement quoted by Reuters (2023), described the situation as "a serious violation of privacy," noting that people walking past parked Teslas were filmed without their knowledge. This is not an incidental feature of Tesla's product design; it is an inherent consequence of building a machine learning training pipeline that depends on continuous, fleet-wide video capture from customer vehicles in real-world environments. Computing professionals designing such systems carry an affirmative obligation to anticipate and mitigate these privacy externalities — an obligation that Tesla's internal culture, as documented by Reuters, did not honour.

3. Problem statement

3.1 How It Should Work: Ideal Standard for Safety-Critical AI

A Level 2 ADAS system operating to professional and ethical standards would fulfil three baseline requirements. *First*, its marketing, naming, and interface would accurately represent its actual capability: it would be named in a manner that reinforces rather than undermines driver

responsibility, and its interface would provide continuous, meaningful reminders that the driver remains the operator. *Second*, its driver monitoring system would be substantive rather than circumventable: a camera-based driver attention monitoring system — standard on many competitor ADAS platforms — cannot be defeated by a coin. *Third*, its release process would be governed by independent safety validation rather than commercial timelines: updates that change how the system handles safety-critical scenarios would undergo regulatory review before deployment to a public fleet.

Beyond these technical standards, an ethical standard also governs what computing professionals owe to the public. The IEEE Code of Ethics (2020) requires engineers to "hold paramount the safety, health, and welfare of the public." The ACS Code of Professional Conduct (2023) extends this obligation to honesty: professionals must not misrepresent the capabilities of the systems they build. When a product is named "Full Self-Driving," both of these obligations are violated simultaneously — public safety is compromised by the complacency the name induces, and honesty is violated by the gap between name and reality.

3.2 What Went Wrong: Ethical Problems Identified

Three distinct ethical problems are identifiable in Tesla's case, each involving different professional actors and different forms of harm.

Epistemic dishonesty and informed consent. Tesla's naming of "Full Self-Driving" constitutes a form of epistemic dishonesty: the company represented its product as possessing capabilities it knew it did not have. This is not a matter of enthusiastic marketing hyperbole. FSD is a USD 15,000 purchase. Buyers making that purchase decision relied on representations Tesla made about what the system could do. O'Dowd (2022) documented a long list of basic safety behaviours — obeying school speed limits, stopping for school buses, reliably recognising pedestrians in crosswalks — that FSD failed to perform at the time of commercial availability. Tesla's own internal data, accessed during NHTSA investigations, showed the company was aware of these limitations. Concealing known limitations from paying customers in a safety-critical product is not merely a commercial ethics issue; it is a professional ethics failure by every engineer, product manager, and executive who approved the product's release under that name.

Design-induced complacency. The torque-based steering wheel sensor is the core of the system's safety architecture. This is compounded by the absence of a cabin-facing camera for genuine driver attention monitoring in earlier hardware generations. The result was a system that communicated "I am watching whether you're paying attention" while possessing no meaningful ability to do so. Computing professionals designing safety-critical systems have a responsibility to anticipate misuse scenarios and design against them; the documented workarounds for Autopilot's monitoring system suggest this responsibility was not met.

Externalised harm to non-consenting third parties. The harms produced by Tesla's accountability gap fell disproportionately on people who had no relationship with Tesla and no opportunity to evaluate or refuse the risk. Jeffrey Nissen, Walter Huang's family members, motorcyclists, pedestrians, and emergency responders who featured in the 16 fatal crashes involving vehicles in Autopilot-related NHTSA investigations (Gigazine, 2024) did not purchase a Tesla. They did not agree to participate in a public beta test. This externalisation of risk — the transfer of consequences from those who made product decisions to those who had no voice in them — is the most serious ethical dimension of the case. It is also the one most directly relevant to computing professionals: when they deploy AI systems into shared public environments, they assume a duty of care to everyone in that environment, not only to the paying customer.

3.3 Why It Failed: Root Causes and Cascading Effects

The root causes of Tesla's accountability gap are structural, not incidental. They originate in the intersection of commercial incentives, institutional culture, and the absence of effective external governance.

The commercial incentive structure is the *first* and most important causal factor. FSD is Tesla's highest-margin software product; it represents a significant revenue stream. Tesla's CEO publicly committed to full autonomy timelines that compressed internal development schedules. This created an organisational environment in which the engineers who raised safety concerns, were overruled by commercial and reputational pressures. The pattern documented by The Washington Post (2023) — engineers advise caution, management overrules, crashes follow — is the canonical form of what professional ethics literature describes as "practitioner capture": professionals who understand a system's limitations being overridden by institutional actors who are insulated from its consequences.

The *second* root cause is the inadequacy of the regulatory framework. NHTSA took approximately three years from the first documented emergency vehicle crashes — in which Autopilot-engaged Teslas struck stationary emergency vehicles on the highway — before opening a formal engineering analysis (Gigazine, 2024). This regulatory latency created a permissive environment in which Tesla could continue deploying updates to a public fleet without meaningful external oversight. The gap between the pace of AI deployment and the pace of regulatory adaptation is not unique to Tesla, but Tesla exploited it more aggressively than most. The 2023 recall of 362,000 FSD Beta vehicles (CNET, 2023) was resolved by an over-the-air update — a mechanism that, while convenient, treats safety corrections as software patches rather than as the formal safety interventions they are.

The *third* root cause is the professional culture embedded within Tesla's engineering teams. De Winter, J.C.F., et al. (2023) finding that the torque sensor could be defeated with a commercial clip-on weight suggests that the engineering team either did not anticipate this failure mode or did not prioritise addressing it. A professional culture that places public safety paramount would have identified the circumventability of the monitoring system as a critical failure mode requiring resolution before commercial deployment; there is no public evidence that Tesla conducted or commissioned such analysis. The cascading effects of these failures extend beyond individual crashes: prolonged exposure to Autopilot may degrade manual driving competence (De Winter, J.C.F., et al., 2023) creating a population of Tesla drivers who are less capable of safe manual operation precisely because of their extended use of a system that was supposed to assist them.

3.4 Stakeholder Analysis and Competing Ethical Considerations

A complete ethical analysis of Tesla's case requires attending to the full range of stakeholders involved, whose interests and ethical positions are not always aligned.

Tesla drivers purchased a product on the basis of representations that proved misleading. Their complacency was, as Chu and Liu (2023) argue, enabled by the system's design. However, they also bear residual responsibility as licensed operators of vehicles in public spaces: the SAE Level 2 classification is publicly documented, and Tesla's user manual explicitly requires supervision. The ethical tension here is between consumer protection principles (which hold that sellers must

not mislead buyers) and individual responsibility principles (which hold that operators of dangerous equipment must remain attentive).

Third parties — pedestrians, cyclists, motorcyclists, and emergency responders — are the most ethically vulnerable stakeholders. They had no contractual relationship with Tesla, no access to its safety data, and no mechanism to express preferences about whether autonomous driving technology should be tested in their environments. From a utilitarian perspective, the aggregate harm imposed on non-consenting third parties by an insufficiently validated technology is difficult to justify by the aggregate benefit to Tesla's customers.

Tesla's engineers and computing professionals occupy the most complex ethical position. Many will have recognised, as the radar removal episode illustrates, that commercial pressures were overriding safety considerations. Professional ethics frameworks are clear that practitioners have an obligation to escalate safety concerns and, if they are consistently overruled, to refuse participation or to alert relevant authorities. The extent to which Tesla's engineers fulfilled these obligations is not publicly documented, but the pattern of product decisions suggests that institutional pressures were sufficient to suppress effective dissent.

Regulators share some of the accountability for this gap. A regulator that takes three years to open a formal probe into a pattern of emergency vehicle crashes is not fulfilling its mandate to protect public safety. This is not primarily a criticism of individual regulators; it reflects the genuine institutional challenge of regulating a technology whose development pace exceeds the pace at which regulatory frameworks can adapt.

4. Conclusion

This investigation has demonstrated that the ethical failures surrounding Tesla's Autopilot and FSD systems are not reducible to technological immaturity. The technology was imperfect — all emerging technology is — but the deaths that resulted were substantially attributable to identifiable professional decisions: naming a supervised system "Full Self-Driving," deploying it without sufficient driver monitoring infrastructure, removing radar against internal engineering advice, and distributing safety-critical updates without formal regulatory oversight. These were choices made by computing professionals operating within an institutional culture that consistently subordinated safety to commercial ambition.

Three findings emerge with particular clarity. *First*, naming is not merely a marketing question; it is an ethical question. When a safety-critical system is given a name that activates user mental models of full automation, the resulting complacency is a foreseeable and designed outcome. The computing professionals responsible for that naming decision bear ethical responsibility for its consequences. *Second*, design choices that permit the circumvention of safety monitoring systems are not neutral technical decisions; they are ethical failures. A torque sensor that can be defeated with a coin is not a driver monitoring system; it is the appearance of one. *Third*, the harm produced by AI systems deployed in shared public environments is borne disproportionately by people who have no voice in their development. Computing professionals have an obligation to those people that is prior to and independent of their obligations to their employer.

The significance of this case for the computing profession extends beyond Tesla. As AI systems are increasingly deployed in safety-critical domains, the structural dynamics that produced Tesla's accountability gap will recur. Commercial incentives will push for premature deployment; regulatory frameworks will lag; internal dissent will be suppressed. The professional response to these pressures is not passive compliance. The IEEE and ACS codes exist precisely for these scenarios: to provide practitioners with a principled basis for escalating safety concerns, refusing to endorse misleading product representations, and, if necessary, acting in the public interest even against institutional pressure.

Looking forward, Tesla is currently pursuing fully driverless robotaxi operations — vehicles with no steering wheel and no human operator. The accountability gap this report has documented will, in that context, become acute: there will be no driver to assign partial responsibility to. The computing professionals who build those systems will bear the full weight of the professional obligations that Tesla's employees have, on the evidence reviewed here, imperfectly honoured. The lessons of Autopilot and FSD are, in this sense, not historical cautionary tales but live professional obligations for every practitioner working in AI-adjacent engineering today.

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